

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1636

George Neal South 230 kV
September 27, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1636. The Interconnection Customer proposes a 100.6 MW wind facility interconnecting at the George Neal South 230 kV substation in the Ohio area. The request was received on December 26, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 435 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$161,679,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mcgill Substation – S Hollywood 100kV	Post Rock – Myrtle Pt 230kV	171.6	135	5.6
Thrive Fontana – Mcmillan 100kV	Mcmillan – Thrive Fontana 100kV	158.5	153	5.5
Thrive Fontana – Mcmillan 100kV Line 2	Mcmillan – Thrive Fontana 100kV Cir. 1	158.5	153	5.5
Mcmillan – Myrtle Pt 230kV L	Post Rock – Myrtle Pt 230kV L	143.2	524	27.3
Mcmillan 230/100kV TX Unit 1	Post Rock – Myrtle Pt 230kV	121.0	219	5.7
Mcmillan 230/100kV Transformer Bank	Post Rock – Myrtle Pt 230kV	121.0	219	5.7
Mcmillan 230 kV/100 kV TX	Post Rock – Myrtle Point Substation 230kV	121.0	219	5.7
Eastown Road – Moreno Tap Substation 230kV	George Neal South – Myrtle Point 230kV	111.6	1,345	100.0
Newport Tie 230/100kV Auto Bank 1	Canal Road – Newport Tie Substation 230kV L	109.4	280	5.3
Newport Tie 230 kV/100 kV XFMR Unit 2	Newport Tie – Canal Road 230kV	109.4	280	5.3
Newport Tie 230/100kV Transformer Unit 3	Canal Road – Newport Tie 230kV	109.4	280	5.3
Eastown Rd 100/1kV TX	George Neal South – Myrtle Pt 230kV	109.3	213	8.1
Denmark – Thrive Fontana 100kV Line	Post Rock – Newport Tie 230kV	104.7	232	6.5
Kampsville – Zona Libre 6007 230kV	George Neal South – Myrtle Pt 230kV	102.6	1,506	100.0
Moreno Tap – Kampsville 230kV L	Myrtle Pt – George Neal South 230kV	100.4	1,523	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Hollywood – McGill 100kV	Rebuild line with higher-capacity conductor	\$4,392,000	100% (\$4,392,000)
Mcmillan – Thrive Fontana Substation 100kV Circuit 1	Rebuild line with higher-capacity conductor	\$4,898,000	39% (\$1,917,000)
Mcmillan – Thrive Fontana 100kV	Rebuild line with higher-capacity conductor	\$3,433,000	39% (\$1,344,000)
Myrtle Pont – Mcmillan 230kV L	Rebuild line with higher-capacity conductor	\$12,810,000	100% (\$12,810,000)
Mcmillan 230/100kV TX Unit 1	Replace transformer with higher-rated unit	\$7,811,000	100% (\$7,811,000)
Mcmillan 230/100kV Transformer Bank	Replace transformer with higher-rated unit	\$11,776,000	100% (\$11,776,000)
Mcmillan 230/100kV TX Bank 3	Replace transformer with higher-rated unit	\$9,611,000	100% (\$9,611,000)
Eastown Road – Moreno Tap 230kV	Rebuild line with higher-capacity conductor	\$82,547,000	100% (\$82,547,000)
Newport Tie 230 kV/100 kV Transformer Bank Bank 1	Replace transformer with higher-rated unit	\$11,036,000	11% (\$1,204,000)
Newport Tie 230 kV/100 kV XFMR Unit 2	Replace transformer with higher-rated unit	\$8,254,000	11% (\$900,000)
Newport Tie 230/100kV Transformer Unit 3	Replace transformer with higher-rated unit	\$8,594,000	11% (\$937,000)
Eastown Road 100/1kV Transformer	Replace transformer with higher-rated unit	\$1,761,000	16% (\$278,000)
Denmark – Thrive Fontana 100kV Line	Reconductor line	\$1,514,000	100% (\$1,514,000)
Kampsville – Zona Libre 6007 230kV	Reconductor line	\$9,148,000	100% (\$9,148,000)
Moreno Tap – Kampsville 230kV L	Reconductor line	\$15,490,000	100% (\$15,490,000)
	Total allocated to project		\$161,679,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.